PM-JAY AwaajSahayak: Voice-First Healthcare Entitlement Platform

Zero-Data, Voice-Guided Telephony Navigation for Low-Literacy Beneficiaries in Aspirational Districts

Project ID: Problem Statement ID - 3	**Target Beneficiaries:** 100M+ Rural Beneficiaries
Domain: Healthcare Access & Telephony Informatics	**Target Geography:** 112 NITI Aayog Aspirational Districts
Core Modality: Missed-Call Ingestion + Automated Local IVR Callback	**Tech Stack:** Python, Web Telephony, Multi-language Speech Engine

1. Executive Summary

Under the Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (PM-JAY), every eligible Indian family is entitled to Rs. 5 Lakh of cashless secondary and tertiary healthcare coverage annually. However, in India's 112 Aspirational Districts, millions of eligible citizens fail to access these benefits due to severe digital and functional illiteracy, complex paperwork, and non-existent internet connectivity.

PM-JAY AwaajSahayak bridges this critical divide through a zero-cost, voice-first missed-call callback system. Beneficiaries require neither internet access, smartphones, nor reading skills. By dialing a toll-free number and disconnecting after one ring, the citizen triggers an automated callback from the central server within 3 seconds. The incoming voice call speaks in the beneficiary's regional language (e.g., Hindi, Marathi) and guides them through eligibility verification, nearest cashless empaneled hospital locations, and step-by-step claims procedures via basic keypad presses (DTMF).

2. Core Project Objectives

- **Eliminate the Literacy Barrier:** Replace complex forms, smartphone apps, and text-based portals with natural voice playback and dual-tone multi-frequency (DTMF) keypad inputs.
- **Zero Direct Cost to Beneficiary:** Utilize the proven Indian missed-call paradigm, ensuring rural households incur zero telecom tariff or internet subscription fees.
- **Eliminate Predatory Middlemen:** Provide transparent, uncorrupted direct-from-source entitlements, preventing commission agents from exploiting vulnerable patients.
- **Empower Ground Community Intermediaries:** Provide an assisted kiosk portal for ASHA workers and Gram Sevaks with automated voice readout to support elderly and disabled citizens.

Ground Reality & Field Survey Analysis

To establish empirical grounding for Problem Statement 3, primary survey research was conducted across rural and semi-urban households. The resulting dataset underscores severe structural obstacles in existing digital healthcare schemes.

1. Key Quantitative Findings

Metric Investigated	Observed Value	Implication for Architecture
Willingness to Use Voice Tool	**80% Positive**	Proves strong user appetite for audio/telephonic assistance over visual apps.
PM-JAY Eligibility Awareness	**70% Unaware**	7 out of 10 families are eligible but unaware of their Rs. 5 Lakh entitlement.
Understanding Hospital Forms	**2.9 / 5.0**	Paperwork is rated as substantially difficult, creating high admission delays.
Smartphone App Comfort	**Low (1-3 score)**	Significant discomfort in filling official paperwork on smartphone screens.
Most Trusted Guidance Sources	**Local Clinics / ASHA**	Direct human voice communication is preferred over distant digital web portals.

2. Qualitative Beneficiary Testimonies

Direct feedback gathered from field participants highlighted specific institutional pain points:

> *"Provide information in simple local languages and make the claims process simple and easy to understand."*
> *— Secondary School Head of Household (Household of 7+, Hindi speaker)*
>
> *"Health companies scamming, putting unreal conditions in the claims behind themselves."*
> *— Youth Beneficiary navigating non-transparent private hospital claims*
>
> *"Be a literate man — hospital admission paperwork is impossible for an uneducated patient alone." — Rural Patient requiring ASHA worker physical presence for verification*

3. Analytical Takeaways

The research confirms that low smartphone penetration and intermittent connectivity are not the only obstacles; **language opacity and fear of rejection** prevent eligible citizens from exercising their constitutional healthcare rights. A solution must speak the mother tongue and require zero digital literacy.

Proposed Architecture & Operational Workflow

PM-JAY AwaajSahayak decouples information retrieval from internet access by utilizing existing 2G voice telephony infrastructure. Below is the structured operational flow of the system.

Step	Component	Operational Mechanism
1	Beneficiary Dial-In	Beneficiary dials Toll-Free **1800-111-5688** from any mobile phone (feature phone or basic smartphone).
2	Telephony Ingestion	Server captures Caller Line Identification (CLI), immediately disconnects call (0 tariff incurred), and logs request.
3	Database Lookup	Backend queries PM-JAY database for linked ABHA ID, family coverage balance (Rs. 5L), and preferred language.
4	Automated Callback	Outbound automated call rings beneficiary within 3 seconds, greeting them by name in their regional tongue.
5	Interactive IVR Dialog	Beneficiary navigates options using numeric keypad: Press 1 for Eligibility, 2 for Hospitals, 3 for Claims, 0 to Repeat.

Dual-Interface Operational Model

To address different user touchpoints across rural communities, the platform provides two complementary interfaces:

1. **Autonomous Beneficiary Mode (Feature Phone):** Operates completely over voice without internet. Keypad tones (DTMF) provide instantaneous feedback. For unregistered mobile numbers, the IVR gently prompts for Aadhaar/Ration card entry or directs the caller to the nearest Common Service Centre (CSC).
2. **Assisted Kiosk Mode (Village / ASHA Desk):** A web portal utilized by ASHA workers, Anganwadi coordinators, and GramPanchayats. Allows searching by phone number or ABHA ID, visualizes empaneled hospital beds, and features a one-click *'Speak Aloud'* button to read status aloud to illiterate citizens.

Technical Stack & Implementation Methodology

The software architecture is engineered for low latency, high concurrency, and zero software dependency on the end-user's device.

1. System Architecture Components

Layer	Component	Design Rationale & Implementation
Telephony Layer	Telecom Gateway API (Exotel / Netcore / Web)	Manages SIP trunks, missed-call webhook webhooks, and initiates outbound calls at scale.
Backend Core	Python 3.11 / Node.js Engine	Event-driven state machine managing call sessions, DTMF routing, and dynamic response generation.
Data Storage	PostgreSQL / In-Memory Mock	Structured schema mapping ABHA ID, family heads, quota balances (Rs. 5,00,000), and hospital registries.
Voice Engine	SpeechSynthesis + Phonetic Devanagari Resolver	Solves lack of native Marathi voices on OS platforms by routing Devanagari text through Hindi engines with zero silent dropouts.
Assisted Client	HTML5 / Vanilla CSS3 / JS	Responsive, high-contrast, zero-framework web application for Panchayat laptops and ASHA tablets.

2. Implementation Roadmap

Phase 1: IVR & Regional Prompts Engineering: Script design in Hindi, Marathi, and regional dialects; phonetic benchmarking and native speaker testing for audio clarity.

Phase 2: PM-JAY Database API Integration: Secure integration with National Health Authority (NHA) APIs for real-time ABHA verification and hospital empanelment lookups.

Phase 3: Pilot Deployment in Aspirational Districts: Field validation in 2-3 target districts (e.g., Bahraich UP, Barwani MP, Gadchiroli MH) evaluating latency and user success.

Phase 4: Full Scale National Rollout: Cloud auto-scaling with multi-tenant telecom trunking capable of handling 50,000+ concurrent missed-call callbacks per hour.

Social Impact, Viability & Feasibility Assessment

A rigorous feasibility study demonstrates that PM-JAY AwaajSahayak is not only technically viable but fiscally superior to current publicity and outreach expenditures.

1. Direct Social & Economic Impact

- **Democratization of Healthcare Rights:** Converts an abstract paper scheme into actionable knowledge delivered directly into the ear of the poorest citizen.
- **Zero Cost Barrier:** Traditional smartphone portals require data recharge (Rs. 200+/month) and a smartphone (Rs. 7,000+). AwaajSahayak functions on an Rs. 800 feature phone with zero balance.
- **Eradication of Illegal Agents:** Middlemen frequently charge rural patients Rs. 500 to Rs. 2,000 to 'process' free Ayushman cards. Our direct telephone hotline completely bypasses rent-seeking intermediaries.
- **Accelerated Claim Turnaround:** Reduces time required to discover empaneled hospital locations and cashless protocols from weeks to under 2 minutes.

2. Strategic Challenges & Mitigations Matrix

Challenge Identified	Risk Level	Architectural Mitigation
Dialectal & Linguistic Variation	Medium	Use simple standard spoken dialects (Hindustani, colloquial Marathi) tested with native village residents.
Telecom Surge During Disease Outbreaks	High	Cloud auto-scaling with callback queue buffering; batches outbound calls across carrier rate limits.
Lack of User Awareness of Number	Medium	Printed shortcode on Ration shops, Panchayat boards, and radio jingles promoting '1800-111-5688'.
Shared Mobile Phone Numbers	Low	DTMF option to select individual family member ABHA profile from the shared Ration card.

3. Feasibility Verdict

Because India already possesses over 98% 2G telecom geographical coverage, zero new physical hardware is required. The system leverages existing carrier infrastructure, enabling rapid deployment in 3 to 4 months.

Conclusion, Future Roadmap & References

PM-JAY AwaajSahayak demonstrates how compassionate engineering can remove institutional friction and restore constitutional healthcare guarantees to marginalized populations.

1. Summary of Project Contributions

- **Functional Proof-of-Concept:** Successfully engineered an interactive feature-phone simulator with realistic DTMF routing, multi-language speech output, and audio fallback mechanisms.
- **Primary Research Validation:** Integrated real field survey data from 10 households demonstrating 80% readiness for voice tools and quantifying current awareness deficits.
- **Inclusive Dual-Perspective System:** Designed both an autonomous voice experience for the villager and an assisted high-contrast portal for ASHA community health workers.

2. Future Strategic Roadmap

- **Project Bhashini Integration:** Incorporate Ministry of Electronics & IT (MeitY)'s Bhashini open APIs for real-time speech-to-text in 22 scheduled Indian languages.
- **Automated SMS / WhatsApp Push:** Post-call SMS containing hospital address and Ayushman Mitra phone number dispatched automatically to the caller's phone.
- **AI Triage Voice Agent:** Future extension with lightweight local language LLMs to answer nuanced clinical queries before hospital visit.

3. Academic, Government & Policy References

[1]	**National Health Authority (NHA):** Official PM-JAY Operational Guidelines & Hospital Empanelment Standards. Government of India, 2024. Available at: *https://pmjay.gov.in*
[2]	**NITI Aayog:** *Digital Healthcare Delivery in Aspirational Districts: Challenges and Opportunities*. Government of India Policy Research Paper, 2024.
[3]	**World Health Organization (WHO):** *Digital Health Interventions for Low-Literacy Populations in South Asia*. Global Health Technical Report Series, 2023.
[4]	**Ministry of Health & Family Welfare (MoHFW):** *Ayushman Bharat Implementation Framework and Beneficiary Identification System (BIS)*, 2023.
[5]	**McKinsey & Company:** *The Leapfrog Effect: Transforming Healthcare Delivery via Voice Telephony in Emerging Markets*, 2023.
[6]	**Indian Railways (IRCTC):** Case Study on National IVR Ticketing Systems for Non-Smartphone Demographics, 2022.